



Fig. 3: Tektronix's Elemental-MS05 oscilloscope (Credit: www.convergent-india.com)



Fig. 4: Full SiC module (Credit: www.rohm.com)



Fig. 5: SiC Trench-MOS (Credit: www.rohm.com)

nents for monitoring, processing and quality control.

T&M equipment for power electronics

Nitin Shetty, chief executive officer, Convergent Technologies, says, "A galvanically-isolated solution like Tektronix TIVM Series IsoVu can resolve high-bandwidth differential signals up to 2500Vpk in the presence of large common-mode voltages. The latest T&M technology for power electronics is 8-channel, 12-bit ADC and deep memory MSO 5 series oscilloscope.

"MSO is available with eight high-resolution channels, easy-to-use user interface and a large HD touchscreen display along with advanced IsoVu probing solutions. It is suitable for designing an inverter, optimising a power supply and testing communica-



Fig. 6: Multimeter (Credit: www.testo.com)

tion links. It measures across a current shunt resistor, debugging electromagnetic interference or electrostatic discharge issues, and eliminates ground loops in a test setup."

Convergent Technologies offers multiple solutions. These include:

- 4-, 6- or 8-flexi channel, 12-bit resolution Tektronix MS05 oscilloscope with advanced power analysis solutions for three-phase power electronics, automotive electronics, power supply design and DC-to-DC power converters
- Battery-operated Tektronix TPS2000 series oscilloscope with fully-isolated and floating channels
- Single- and three-phase power analysers: PA3000 and PA1000 series from Tektronix

Vijay Bolloju, manager - applications engineering, Rohm Semiconductors, says, "The requirement of power systems for automotive systems have increased considerably due to heavy electric vehicles and their development activities. This necessitates low-voltage and high-current power supplies to characterise system performance. Efficiency, total harmonic distortion and power factor correction of the systems are measured using regulated AC and DC supplies and power analysers."

He adds, "We provide SiC MOSFET technology and full SiC modules, an extensive series of SiC products like MOSFETS, Schottky barrier diodes and more. We also have regulated AC and DC power supplies, electronic

loads, LED load simulators, high-bandwidth oscilloscopes with math functions and spectrum analyser packages. Torque simulators such as Magtrol are used for testing motor drives. Low-footprint, precise current probes are used for measuring the currents in circuits."

Parag Yelegaonkar, business development manager, Testo India, says, "Our latest and most advanced solutions are electrical instruments for inspecting electrical components and circuit failure threats. These ensure longevity and efficiency of various power electronic components that control and regulate the flow

of electrical energy.

"Our unique clamp meters and multimeters are intuitive, can measure several parameters in a single component and have patented features. The new range of thermal imagers with smartphone integration are designed to deliver networked thermography mostly used for predictive and preventive maintenance in electric and power sectors. Wireless operation, with the ease of saving and transferring data over networks, makes these user-friendly, interactive and cost-effective."

Challenges

Measuring current is always a challenge, as the current probes tend to be bulky and invasive. High-bandwidth compact current probes measure low currents (< 5A). These are needed to test low-power systems like ceiling fans and other consumer appliances.

To test high-speed power devices such as SiC and GaN MOSFETs, higher bandwidth measuring equipment are needed to capture fast-rising switching waveforms.

Equipment for measuring electromagnetic compatibility and surge tests tend to be quite expensive. Low-cost pre-compliance test equipment can expedite the design process.

Power quality, switching loss, harmonics, ripple, modulation and safe operating area measurements are other challenges in power electronics. These can be overcome by using latest and faster testing devices. **EFY**